

DFT methods used, or a sign of under-convergence for several very similar-in-energy polymorphs. As such, it should not be seen as a failure of the model. Likewise, TlNO_3 thallous nitrate is experimentally-known in the Pnma space group [102] with a calculated (PBE) band gap of 2.8 eV [103], however the generated crystal appears quite different to the experimental structure due to a change in Tl co-ordination around NO_3^- anions.

Bulk modulus conditional generation

For the high bulk modulus generation task, a manual review of ten generated crystals chosen at random was performed, as well as the two representative structures with high bulk modulus that were highlighted in Fig. 5(f). All structures show a strong compositional bias, containing a mix of refractory elements Re, W, Mo and Ir and frequently also B and C; this is consistent with literature on superhard materials (which includes materials with high bulk modulus). When the composition contains only refractory elements, the generated structure typically seems like an ordered approximation of an alloy of that composition, while those that also contain B or C typically take a very anisotropic, layered structure. As an example, Re_3Ir highlighted in Fig. 5 can be interpreted as an ordered approximation of a $\text{Ir}_x\text{Re}_{1-x}$ alloy, which has been previously synthesized and is known to exist in solid solution [104]. The $\text{Re}_3\text{B}_2\text{C}$ example is more unusual, with layers of all Re, B, Re, C, Re, ..., and nominally can charge balance. We note that the bulk modulus calculated is an averaged quantity over the full elastic tensor, and we have not examined the directional bulk moduli in these highly anisotropic systems.

D.7 Designing low-supply-chain risk magnets

This section provides supplementary information to the results in Section 2.6.

D.7.1 Additional experimental details

To generate structures conditioned on magnetic density and HHI score, we fine-tune our base model on these two properties, encoded as in Appendix D.6. To evaluate the performance of our model, we proceed as detailed in Appendix D.6, and generate 512 samples with our fine-tuned model, by conditioning on a magnetic density value of $0.2 \text{ \AA}^{-3}$ and an HHI score of 1200. Of those, 130 samples remain after filtering by stability and uniqueness following the DFT relaxation. Finally, a total of 112 structures pass the novelty check w.r.t. the reference dataset and are reported in Fig. 6(a).

D.7.2 Additional qualitative analysis of structures

Targeting a low HHI index in addition to high magnetic density steers MatterGen away from generating structures with Co, which is associated with poor HHI scores. Example structures include $\text{Fe}_x\text{Mn}_{1-x}\text{O}$ rocksalt alloys (MnFe_3O_4 , MnFe_8O_9); however, they only exhibit a high magnetization density in a hypothetical ferromagnetic state, and not in their actual antiferromagnetic ground state. Other similar example outputs include a defected FeO containing vacancies (Fe_8O_9), and a body-centered-cubic Fe_8Au system, both of which are well-known experimentally. While the joint optimization task

could further be extended to produce more reasonable candidates—e.g., by penalizing expensive elements, and preferring metallic systems more likely to be ferromagnetic—the overall performance of the model with respect to the labels used for training is reasonable. It is possible that a better treatment of alloy systems would be required to improve performance of the high magnetic density generation task, and to ensure that the generated structures are truly novel.

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